An Ancient Ocean of Giant Predators Reveals a Surprising Twist in Evolution

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Long ago, in an ocean that covered what is now central Colombia, giant marine reptiles ruled the waves. Long-necked **plesiosaurs** cruised beside marine crocodile relatives, turtles, sharks and **crustaceans**. Now **fossils** from that lost world are changing how scientists think about evolution.

Researchers from McGill University and the Centro de Investigaciones Paleontológicas in Colombia studied **fossils** from the Paja Formation, near Villa de Leyva. The rocks, from the Early Cretaceous period, preserve an unusually detailed picture of an ancient ocean community the scientists call the Paja Biota, one of the most ecologically complex marine communities found so far.

What makes it special is who was sharing the sea. Ancient animal groups from the Jurassic period, already heading toward extinction, were living side by side with newer Cretaceous groups that were just beginning to flourish. In an earlier study, the team found the food web was exceptionally intricate, with an unusually large variety of top **predators** at the highest levels.

McGill biology professor Hans Larsson and PhD graduate Dirley Cortés have given this pattern a new name: a '**Biotic Inflection**.' It describes an evolutionary turning point when older groups nearing extinction coexist with newer ones starting to expand and diversify.

The researchers think a 'perfect storm' of conditions made it possible: warm temperatures, high sea levels and nutrient-rich coastal waters, plus new ocean connections between the eastern Pacific and the early Atlantic through the region that is now Colombia.

The big idea is that major changes in life on Earth do not always need a mass extinction. Sometimes, Larsson said, 'a perfect storm of environmental factors we know to promote ecological complexity coincided with ancient lineages that were about to disappear. At the same time, new groups were beginning to flourish.'

But the researchers stress that more evidence is needed. More **fossils**, better dates and comparisons with rocks elsewhere will show whether this pattern happened in other times and places too. 'I think we've discovered a new model of global biodiversity change,' Larsson said.

Word Treasure

plesiosaurs -- Ancient sea reptiles with long necks that lived in the oceans millions of years ago.

fossils -- The preserved remains of animals or plants that lived long ago.

predators -- Animals that hunt and eat other animals.

crustaceans -- Sea animals with hard shells, like crabs, shrimps and lobsters.

Biotic Inflection -- A turning point in evolution when older groups nearing extinction live beside newer groups starting to grow.

Background you need

Fossils are the preserved remains of animals and plants that lived long ago, and they help scientists rebuild lost worlds.

Evolution is the slow change of living things over millions of years, as new groups appear and old ones die out.

A food web shows who eats whom in a community, from tiny creatures up to the top predators.

A mass extinction is an event in which huge numbers of species die out in a short time, but this study suggests big changes can also happen gradually.

Break it down

LEAD: Fossils from an ancient ocean in Colombia are changing how scientists think about evolution.

KEY EVENT: Old Jurassic animal groups nearing extinction lived side by side with newer Cretaceous groups just beginning to flourish.

+- WHO: Researchers from McGill University and the Centro de Investigaciones Paleontológicas in Colombia.

+- WHEN: The rocks come from the Early Cretaceous period, in the Paja Formation near Villa de Leyva.

EVIDENCE: The food web was exceptionally intricate, with an unusually large variety of top predators.

OPEN QUESTION: Did this pattern happen in other times and places too, or only here?

Why it matters

This discovery changes how scientists think about life's history: big changes in nature do not always need a disaster, and understanding that helps us see how today's living things might change too.

Test what you remember

1. Where were the fossils found?

- (A) A desert in Egypt
- (B) The coast of Australia
- (C) The Paja Formation, near Villa de Leyva in Colombia
- (D) The Amazon rainforest in Brazil

2. Which animals ruled the ancient ocean?

- (A) Giant penguins
- (B) Dinosaurs on land
- (C) Whales and dolphins
- (D) Giant marine reptiles such as plesiosaurs

3. What is special about who shared the sea?

- (A) Only one species lived there
- (B) All the animals were the same age
- (C) Old Jurassic groups lived beside newer Cretaceous groups
- (D) No predators lived there

4. What name did the scientists give the pattern?

- (A) A 'Biotic Inflection'
- (B) A 'Food Web'
- (C) A 'Mass Extinction'
- (D) A 'Perfect Storm'

5. What conditions created the 'perfect storm'?

- (A) A drought and an earthquake
- (B) A meteor strike and a volcano
- (C) Cold temperatures and low sea levels
- (D) Warm temperatures, high sea levels and nutrient-rich waters

6. What do the researchers say is still needed?

- (A) A new museum
- (B) More funding for space travel
- (C) A bigger research team
- (D) More fossils, better dates and comparisons with rocks elsewhere

Answer key (try the quiz first!): 1: C 2: D 3: C 4: A 5: D 6: D

Questions to think about

1. Professor Hans Larsson: He says the team has discovered a new model of global biodiversity change, a 'Biotic Inflection'.

2. The research team: It stresses that more evidence is needed, with more fossils, better dates and comparisons with rocks elsewhere.

MY THOUGHTS (at least 20 words):